

Map Enumeration

A Problem-Solving Exposition

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Conventions

- All the graphs we consider can have loops and multiple edges.
- Unless otherwise stated, our graphs are connected.

Definition 0.1 (Half-Edge [1]). We can think of an edge as a pair of half-edges, matched together. This helps to avoid some difficulties dealing with multiple edges and loops.

Definition 0.2 (Corner [2]). A corner is the angular sector delimited by two consecutive edges around a vertex.

$::: \{\text{\#def-degree of a vertex}\}$

Degree of a Vertex

The degree of a vertex v in a graph G is the number of half-edges incident to v .

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Definition 0.3.